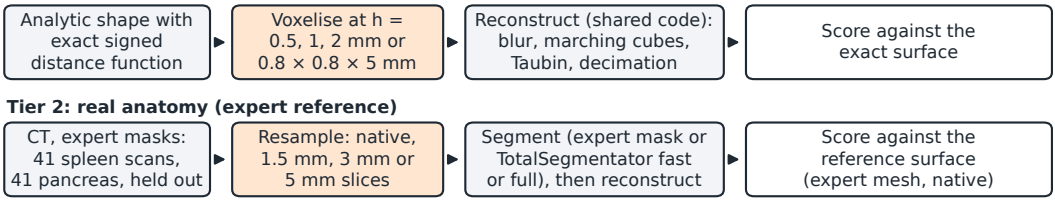
Tier 1: digital phantoms (exact truth)
Analytic shape with exact signed distance function
Voxelise at h = 0.5, 1, 2 mm or 0.8 × 0.8 × 5 mm
Reconstruct (shared code): blur, marching cubes, Taubin, decimation
Score against the exact surface
Tier 2: real anatomy (expert reference)
CT, expert masks: 41 spleen scans, 41 pancreas, held out
Resample: native, 1.5 mm, 3 mm or 5 mm slices
Segment (expert mask or TotalSegmentator fast or full), then reconstruct
Score against the reference surface (expert mesh, native)
sampling error
segmentation and reconstruction error